ECO702 Macroeconomics - Mid-sem exam

February 21, 2025

There are three questions which count for a total of 60 points. There will be points for partial answers so writing *something* is a strictly dominant strategy. You have 2 hours to complete the exam. All the best!

1. **Arrow-Debreu Eq. in a pure exchange economy (20 points)**

   This is a modification of the two person exchange economy we solved in class. Consider a pure exchange economy in which there are two agents who live forever. Time is discrete and indexed by $t = 0, 1, 2, ...$ In each period, the agents trade a non-storable consumption good. Consumer i consumes c_t^i at time t. Agents have homogeneous preferences represented by

   $$u(c^i) = \sum_{t=0}^{\infty} \beta^t ln(c_t^i)$$

   with $\beta \in \left(\frac{\sqrt{5}-1}{2}, 1\right)$. Here c^i denotes the consumption stream of agent $i \in \{1, 2\}$. That is, $c^i = \{c_0^i, c_1^i, c_2^i, ...\}$ Agent i receives the deterministic endowment,

   $$e_t^1 = \{1, 0, 0, 1, 0, 0, 1, 0, 0, 1, ...\}$$
   $$e_t^2 = \{0, 1, 1, 0, 1, 1, 0, 1, 1, 0, ...\}$$

   (a) **(5 points)** Precisely describe an Arrow-Debreu market structure for this economy. Explain when markets are open, who trades with whom, and so on. Define a competitive Arrow-Debreu Equilibrium (ADEq.) for this economy. Use the notation $\hat{p}_t$ for the equilibrium prices and $\hat{c}_t^i$ for the equilibrium consumptions. What is the aggregate endowment in each period?

   (b) **(10 points)** For the ADEq., find the equilibrium prices and consumption streams of each agent. Prove that the equilibrium consumption sequences are constant over time. Use the normalization $\hat{p}_0 = 1$. Graph the sequences of consumptions and prices. Rank the equilibirum discounted utlities $\left\{u(\hat{c}^i)\right\}_{i=1,2}$. Which consumer is relatively better off?

(c) **(5 points)** Suppose that one of the consumers markets a derivative asset that promises to pay 0.05 units of consumption each period. What would be the price of that asset?

2. **Social planner's problem and Arrow-Debreu Eq. (20 points)**

The first welfare theorem was leveraged by Negishi (1960) to propose a convenient way of calculating the set of Arrow-Debreu Equilibrium (ADEq.).[1] The steps are as follows.

(a) Compute all Pareto efficient allocations (i.e. solve a social planner's problem with *arbitrary* weights).

(b) The First Welfare Theorem tells us that all competitive alllocations (i.e. ADEq.) are Pareto efficient. Therefore, by solving for all efficient allocations, we have also solved for the competitive ones.

(c) Use agents' budget constraints (which must bind if marginal utility is strictly positive) to isolate the efficent allocations that are also competitive allocations.

Implement Negishi's method to sovle for the ADEq. allocation in the two person endowment economy without uncertainty that we studied in lecture.

Recall that person 1's (2's) endowment of the consumption good is 2 units at any t even (odd) and 0 otherwise. That is,

$$y_t^1 = 2, 0, 2, 0, ...$$

$$y_t^2 = 0, 2, 0, 2, ...$$

Person i's preferences are given by,

$$\sum_{t=0}^{\infty} \beta^t ln(c_t^i)$$

Hint: Attach Lagrange multipliers equal to $\frac{\theta_t}{2}$ when setting up the planner's problem. It turns out that these shadow prices $\frac{\theta_t}{2}$ from the social planner's problem are equal to the ADEq. prices. You will want to use this equivalence.

3. **First welfare theorem (20 points)**

Complete the steps to prove the first welfare theorem for a deterministic economy. That is, prove the following proposition.

[1] Note that solving for an ADEq. requires one to figure out *prices* in addition to *allocations* which makes things more complicated.

Proposition. *Let* $\left(\{\hat{c}_t^i\}_{t=0}^{\infty}\right)_{i\in I}$ *be an Arrow-Debreu allocation for an endowment economy with endowments given by* $\left(\{y_t^i\}_{t=0}^{\infty}\right)_{i\in I}$ *and associated prices* $\{\hat{p}_t\}_{t=0}^{\infty}$*. Then* $\left(\{\hat{c}_t^i\}_{t=0}^{\infty}\right)_{i\in I}$ *is Pareto Efficient.*

The proof will proceed by contradiction. Assume that $\left(\{\hat{c}_t^i\}_{t=0}^{\infty}\right)_{i\in I}$ is not Pareto Efficient. Then, there exists another feasible allocation $\left(\{\tilde{c}_t^i\}_{t=0}^{\infty}\right)_{i\in I}$ such that

$$U(\tilde{c}^i) \equiv \sum_{t=0}^{\infty} \beta^t u(\tilde{c}_t^i) \geq \sum_{t=0}^{\infty} \beta^t u(\hat{c}_t^i) \equiv U(\hat{c}^i)$$

for all $i \in I$, and,

$$U(\tilde{c}^i) = \sum_{t=0}^{\infty} \beta^t u(\tilde{c}_t^i) > \sum_{t=0}^{\infty} \beta^t u(\hat{c}_t^i) \equiv U(\hat{c}^i)$$

for atleast one i. Without loss of generality, assume that the strict inequality holds for $i = 1$. Also assume that $u(.)$ is strictly increasing (i.e. $u' > 0$).

(a) **(7.5 points)** Show that,

$$\sum_{t=0}^{\infty} \hat{p}_t \tilde{c}_t^1 > \sum_{t=0}^{\infty} \hat{p}_t \hat{c}_t^1$$

(b) **(7.5 points)** For all $i \neq 1$, show that,

$$\sum_{t=0}^{\infty} \hat{p}_t \tilde{c}_t^i \geq \sum_{t=0}^{\infty} \hat{p}_t \hat{c}_t^i$$

(c) **(5 points)** Complete the proof by combining the two intermediary results to find a contradiction.